

Project 9: Precision Tours

Jeric, Sylas, Alvin, Shaishav

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Introduction

Purdue University conducts large scale campus tours for prospective students and their families throughout the academic year. These tours are designed to introduce visitors to campus life, academic programs, facilities, and student experiences. After an initial presentation, visitors are typically organized into smaller tour groups led by student guides.

Campus Tours at Purdue

Visitors experience campus tours in a rather structured sequence:

- Attend an introductory presentation about Purdue.
- Transition to an outdoor or lobby space where tour groups are formed.
- Join a student tour guide and walk through key campus locations.

Each tour guide is responsible for leading a group of around 15–20 visitors, providing general information about campus, and answering questions along the way. These tours are intended to give a broad overview of Purdue, while also allowing visitors to ask questions based on their individual interests.

However, visitors attend tours with very different goals. Some have a clear intended major and want to see specific academic buildings or labs. Others are interested in student life, housing, or identity based experiences, such as what it is like to be a first generation college or international student.

The Problem

The transition from the presentation to tour group assignment can be unclear and loosely structured. Visitors often rely on informal cues such as where they are standing or how lines form, rather than being guided by their interests or needs. As a result, many are placed into mixed groups that may not align with what they want to learn.

Because each guide must address a wide range of questions at once, tours tend to stay general. Visitors with specific academic or personal questions may not get detailed answers, or may feel hesitant to ask in a large group setting. Some leave the tour needing to search online or ask additional questions afterward to fill in gaps.

This project examines the campus tour grouping and assignment experience at Purdue. Our team conducted interviews and initial research to understand how visitors find their tour group, how they are placed, and whether the experience aligns with their expectations. The goal is to identify where confusion and mismatch occur, and to use those insights to inform design improvements that make the tour assignment clearer and more relevant to each visitor.

Contextual Research

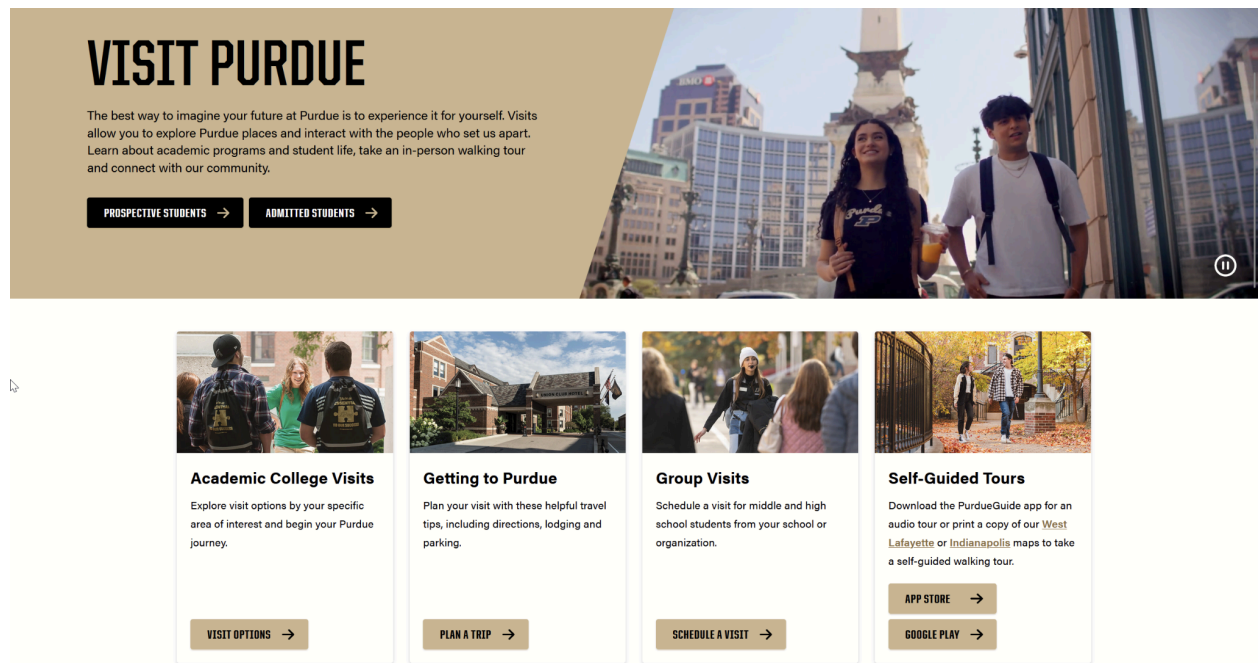
To develop our questions, we broke down the campus tour into some moments where we think that confusion or mismatch is likely to happen. We focused on these parts because visitors have to make quick decisions with limited guidance.

From there, we wrote questions that focus on what people actually do, such as how they find their group, how groups are formed, and whether they feel comfortable asking questions. The goal is to see what is happening and identify where things become unclear or do not match what visitors need.

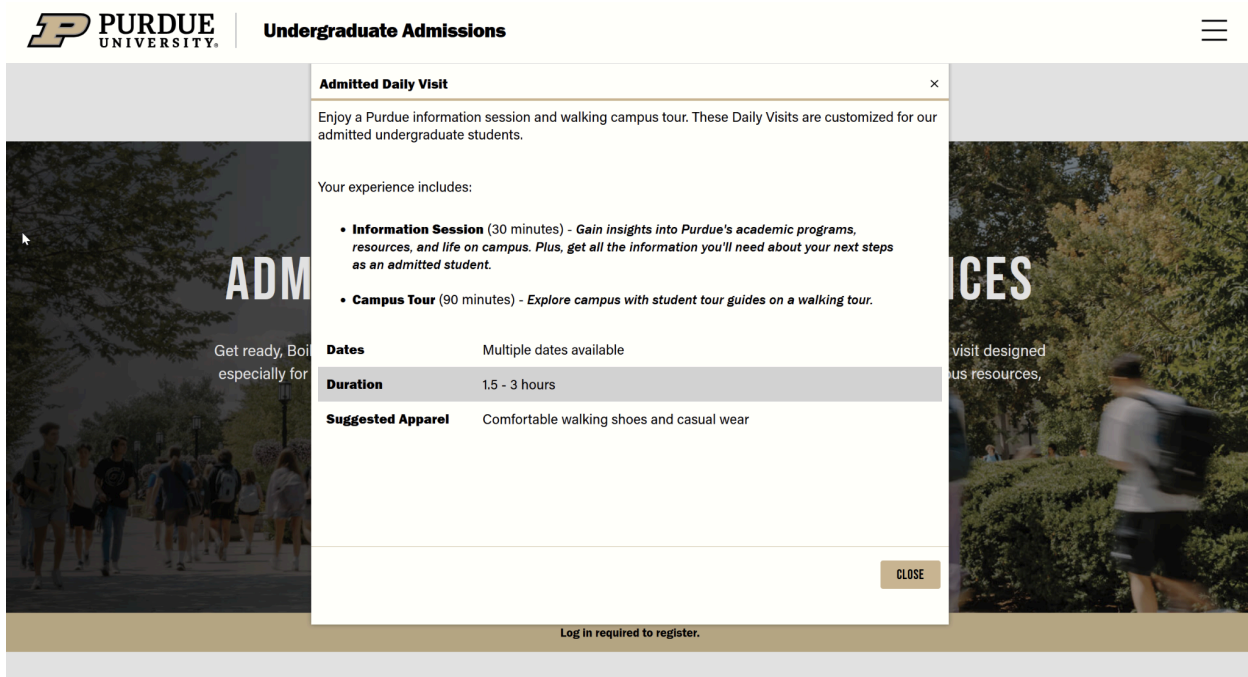
Questions

1. Transition and Way Finding
 - i. What did you do immediately after the presentation ended
 - ii. How did you find your tour group
 - iii. What did you rely on most (staff, signs, other people)
 - iv. Did you feel confused at any point during this transition
 - v. Was anything unclear about where you were supposed to go
 - vi. Did you end up in your group intentionally or just follow others
2. Group assignment + Alignment
 - i. What were you hoping to learn from the tour
 - ii. Did you feel like your group matched your interests
 - iii. Were you grouped based on anything specific or just timing
 - iv. Did you feel like the tour was relevant to you
 - v. Was there anything you wanted to see that wasn't included
3. Group Size + Interaction
 - i. How did you feel about the size of your group
 - ii. Did you feel comfortable asking questions
 - iii. If not, what made it uncomfortable
 - iv. Did you notice others asking questions
 - v. Would you have asked more questions in a smaller group
4. Information Delivery + Unmet Needs
 - i. Did the tour answer your main questions
 - ii. Was the information too general or detailed enough
 - iii. What questions do you still have after the tour
 - iv. Did you need to look up information afterward
 - v. What would have made the tour more helpful for you

Tours (Website Specific)



- The first thoughts we had in regards to the website page was to sort the content better.
- The Academic College Visits and Group Visits panels are both referring to different kinds of touring experiences. Self-guided tours are also different, but they are a category of their own in terms of how they function.
- Getting to Purdue takes you to a page about lodging and travel services that can take you to Purdue if needed. While this is relevant to visiting campus, the issue comes from the odd positioning of this panel in between the two main touring options and the fact that it's not made immediately clear what this page offers.
- A better way to re-organize this page would be to put the two main touring options side by side as the first thing you see, with clearer and better defined titles telling you exactly what each one will lead to.
- The self-guided tours section will be a section of its own, oriented horizontally below, as it does not lead to unique pages like the main two options but instead it simply allows the user to download the Purdue Guide app.
- This is the same reason Getting to Purdue will also be its own section, placed below the Self-Guided Tours panel. Along with that, it'll be named something else so that it's clearer what the section actually contains.
- Each panel will retain their descriptions, which may be improved wherever possible.



Tours

Notes / patterns from interviews

- People come in with different goals. Some want major-specific info like labs, classes, careers. Others care more about dorms, social life, or general vibe.
- Some admitted students are not deciding if they like Purdue, they are deciding where they fit or confirming details.
- Prospective students are more exploratory. They want a broad overview.
- One fixed tour does not match all of these needs.

Transition + Wayfinding

- Most people leave the presentation and look around first
- Many follow the crowd instead of clear instructions
- Signs and staff are not always obvious
- Some are not sure if they are in the right group

Group Assignment + Alignment

- Groups is completely random

- Placement is based on timing or position, not interest
- Tours feel too general for people with specific goals
- Some wanted to see certain buildings but could not

Group Size + Interaction

- Larger groups reduce participation
- Some feel uncomfortable asking questions
- Only a few people speak during the tour
- Smaller groups would make people more willing to ask

Information Delivery + Unmet Needs

- Information is broad and scripted
- Some questions are not answered during the tour
- People look things up afterward
- Admitted students want more detail

Secondary Research

We looked at existing research to understand what visitors expect and where problems tend to show up. This keeps us from relying only on assumptions or a small set of interviews.

- Compare what we saw to known patterns to check if the problem is real at all.
- Turns “groups feel random” into clear issues like assignment, navigation, and mismatch
- Helps us write direct, non leading interview questions (we did this alongside writing the interview questions)
- Explains confusion using system limits and how people behave in crowds
- Backs up our findings so they are not just personal observations

In this project, secondary research lets us treat the campus tour as a system made up of wayfinding, group assignment, and information delivery, so we can better identify where mismatches happen and why.

Questions

After secondary research, we refined our focus into the following questions:

- How participants behave during tours, including whether they ask questions, what prevents them from doing so, and how factors like group size and environment affect engagement?
- How do different grouping strategies (interest-based vs random vs pre-assigned) affect visitor satisfaction, engagement, and perceived relevance of the tour experience?
- How do wayfinding systems (signage, staff guidance, digital tools) impact transition efficiency, confusion rates, and accuracy in group formation after large presentations?
- How does tour structure (fixed route vs flexible or modular routes) influence information retention, perceived personalization, and overall decision confidence?

Results

1. How participants behave during tours, including whether they ask questions, what prevents them from doing so, and how factors like group size and environment affect engagement?
 - a. Participation
 - i. Most people just listen instead of asking questions
 - ii. Only a small number actively engage
 - b. Pressure
 - i. People avoid speaking in front of strangers
 - ii. Fear of interrupting or asking a bad question
 - c. Group size

- i. Larger groups is harder to talk in (quantify large and small is hard)
 - ii. Smaller groups is generally easier to talk in
 - d. Guide behavior
 - i. Lecture style almost always have low engagement
 - ii. Interactive has higher engagement
 - e. Group familiarity
 - i. More likely to talk to family
 - ii. Nobody talks to strangers
 - f. Environmental constraints
 - i. Walking and time pressure make questions feel disruptive
 - ii. Microphone often fails or is inconsistent, making it harder to hear the guide and follow along
 - g. Outcome
 - i. High engagement = small groups, interactive guides, low pressure
 - ii. Low engagement = large groups, lecture style, constant movement
- 2. How do different grouping strategies (interest based vs random vs pre assigned) affect visitor satisfaction, engagement, and perceived relevance of the tour experience?
 - a. Random grouping (current)
 - i. Based on timing or position, not interest
 - ii. Easy to manage at scale
 - iii. Leads to mixed groups with different goals
 - b. Different visitor goals
 - i. Some visitors just want to get a general feel or vibe of the school
 - ii. Others want specific information like majors, buildings, labs, or career paths
 - iii. Admitted students often look for confirmation and details, not a broad overview
 - iv. One fixed tour does not match all of these needs
 - c. Relevance
 - i. Information feels too general
 - ii. Visitors cannot focus on what they care about
 - iii. Specific interests are often not addressed
 - d. Engagement
 - i. Information feels too general for those seeking specifics
 - ii. Visitors cannot focus on what they personally care about
 - iii. Key interests like majors or facilities are often missed
 - e. Impact on satisfaction
 - i. Visitors leave with unanswered questions
 - ii. Some need to look up information afterward
 - iii. Lower sense of fit with the school
 - f. Interest based grouping
 - i. Groups formed around major, goals, or identity

- ii. Higher relevance of information
 - iii. More targeted questions and discussion
 - g. Pre assigned grouping
 - i. Visitors choose or are assigned before arrival
 - ii. Reduces confusion during transition
 - iii. Improves alignment but requires coordination
 - h. Tradeoffs
 - i. Better alignment vs harder logistics
 - ii. Limited number of guides may restrict options
 - i. Outcome
 - i. Higher satisfaction with aligned groups, relevant content, and more interaction
 - ii. Lower satisfaction with random groups, generic tours, and passive experience
3. How do wayfinding systems (signage, staff guidance, digital tools) impact transition efficiency, confusion rates, and accuracy in group formation after large presentations?
- a. Current behavior
 - i. Grouping is completely random, based on who is nearby
 - ii. Staff take around 15 to 20 people at a time from STEW and move them out
 - iii. Visitors do not know who they will be grouped with
 - iv. Most people just follow whoever moves first
 - b. Signage
 - i. Limited role since grouping is not predefined
 - ii. Almost no signage except a stand
 - c. Staff guidance
 - i. Primary way groups are formed
 - ii. Works for moving people quickly, not for clarity or alignment
 - iii. Little explanation given during grouping
 - d. Digital tools
 - i. Not used in current process
 - ii. No support for pre grouping or navigation
 - e. Where confusion happens
 - i. Right after the presentation when people start moving
 - ii. When multiple groups form at once
 - iii. When people are unsure if they are in the right group
 - f. Impact on efficiency
 - i. Fast for staff to organize large numbers
 - ii. Creates short term flow but not clear understanding
 - g. Impact on accuracy
 - i. High chance of being placed in a group that does not match interests
 - ii. People rely on chance rather than intention
 - h. Outcome

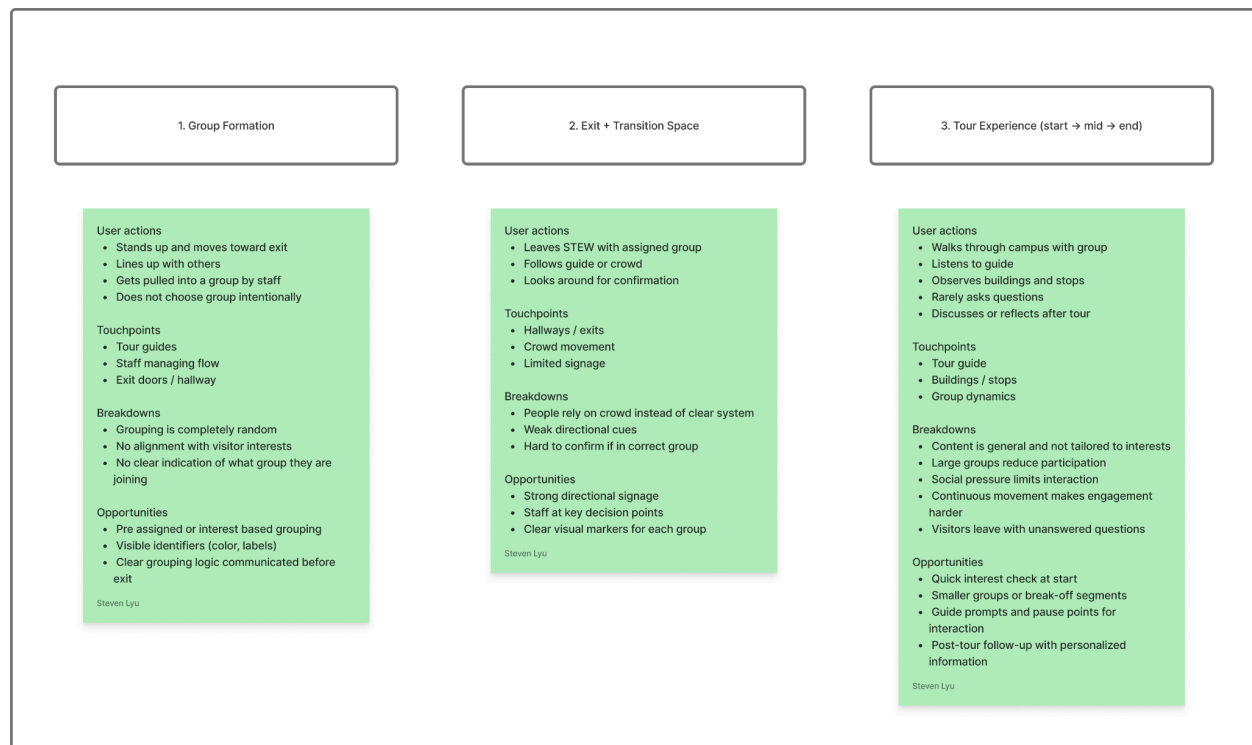
- i. System is efficient for throughput
 - ii. Low clarity and low alignment with visitor needs
- 4. How does tour structure (fixed route vs flexible or modular routes) influence information retention, perceived personalization, and overall decision confidence?
 - a. Fixed route (current)
 - i. Same path for every group, covering standard buildings and talking points
 - ii. Easy to manage and keeps tours consistent
 - iii. Guides stay on a script to keep timing and flow
 - b. Impact on information retention
 - i. Visitors remember general facts but not details tied to their interests
 - ii. Hard to connect information to personal goals
 - iii. Information feels broad and less memorable
 - c. Impact on personalization
 - i. Low sense of personalization
 - ii. Everyone hears the same content regardless of why they came
 - iii. Specific needs like majors, labs, or student life are not fully addressed
 - d. Impact on decision confidence
 - i. Visitors get a surface level understanding
 - ii. Many leave unsure about fit
 - iii. Some need to do additional research afterward
 - e. Flexible or modular routes
 - i. Core route with optional break offs based on interest
 - ii. Ability to adjust stops or depth depending on the group
 - iii. Guides can tailor content to what the group cares about
 - f. Impact on experience (modular/flexible routes)
 - i. More relevant information is easier to remember and connects directly to personal goals
 - ii. Visitors feel the tour matches what they came for
 - iii. Leads to a clearer understanding of fit
 - g. Tradeoffs
 - i. More complex to manage and coordinate
 - ii. Requires more guides or flexible scheduling
 - h. Outcome
 - i. Fixed routes are efficient but generic
 - ii. Flexible routes increase relevance, engagement, and confidence

Synthesis

Expectation	Reality
Tour tailored to their major / interest	Mixed groups, generic script, no relevance to individual goals
Open conversation and easy to ask questions	Large groups and passive listening
Insider perspective from the guide	Curated answers and time constrained responses
Clear directions, smooth transitions, easy group joining	Unclear directions, people following the crowd
See places tied to their future (labs, major specific buildings)	Standardized route, limited access, not aligned with interests
Leave knowing whether they belong there	Hard to judge due to scale, lack of interaction, surface level exposure

Journey Mapping

We then used journey mapping to force everything we collected into one continuous flow. Interviews gave us fragments like “people follow the crowd,” “groups feel random,” “no one asks questions,” but those were isolated observations. Putting them on a timeline exposed how they connect.



Here are some takeaways that we've gathered:

What happens during the tour (like low engagement and general content) traces back to the early decisions, and smaller issues like difficulty hearing the guide due to unreliable microphones or visitors repeatedly asking the same directional questions because of confusing spaces make engagement even harder. The biggest problems are at **group formation** and the **transition space**, and everything after that is just a result of those first steps. Group formation happens right as people line up after the presentation. **Once they are placed into their groups, the mismatch is already locked in, so the tour can't really be fixed later.** Confusion spikes right when people leave and when groups start forming then drops off as people stop deciding and just follow along.

The system moves people quickly but that speed comes at the **cost of clarity and relevance** since people end up in groups that do not match what they want. What happens during the tour, like low engagement and general content, traces back to the early decisions. The problems are at group formation and the transition space, and everything after that is just a result of those first steps.

Breakdown Goals

We broke down goals to make the mismatch clear at each step instead of treating the tour as one general problem. Journey mapping shows where things go wrong, but this step focuses on what users are actually trying to do and where the system fails them. It turns confusion into specific gaps between what users want and what they get.

- **Group formation**
 - Want: group that matches interests
 - Get: random placement, no control
- **Transition**
 - Want: clear direction and confirmation
 - Get: follow the crowd and guess
- **Tour**
 - Want: relevant info and ability to engage
 - Get: general content / low interaction

The differences in impact become obvious after laying everything out. Obviously not every breakdown matters the same since some happen early and shape everything after and others are just outcomes.

Prioritize

- **Group formation (highest)**
 - Happens first
 - Locks in the entire experience
- **Transition (next)**
 - Highest confusion point
 - Leads directly to random grouping
- **Tour (lower)**
 - Issues come from earlier steps
 - Fixing it alone does not solve the core problem

Analysis

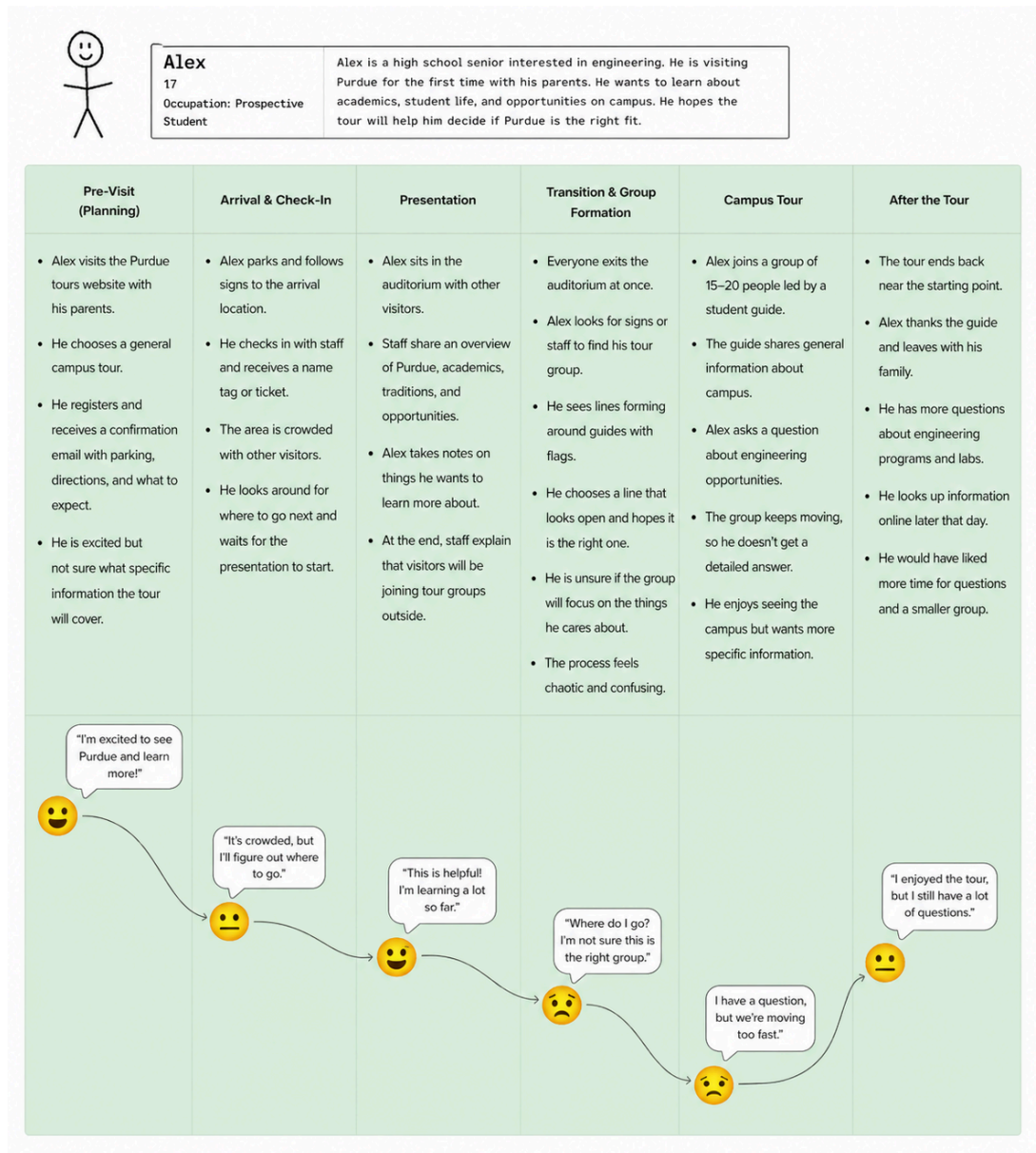
Task Analysis

We broke down the campus tour into the actual steps visitors go through, starting from picking a tour on the website to walking through campus with a guide. On the website, people try to find something that matches what they want, but the structure makes it hard to tell what each option really offers. A lot of them end up picking something without fully understanding the difference.

After the presentation ends, people move into the transition space. This is where things get messy. Most visitors don't make a clear decision about where to go. They pause, look around, and then start following whoever moves first. Staff and signs exist, but they are not strong enough to guide behavior on their own.

Group formation happens right away during this movement. People get pulled into groups based on where they are standing or how the line forms. There is no real moment where they choose a group that matches what they want.

From that point on, the rest of the experience plays out within that group. Whether someone gets useful information or feels comfortable asking questions depends on how well that group matches their goals. By the time the tour starts, most of the important decisions have already been made.



(See Appendix for the JPG of the Journey Map)

Here we created a journey map that lays out the full campus tour experience from the visitor's point of view, moving step by step from pre-visit planning through the tour and after the visit. We used a simple stick figure persona to keep the focus on behavior and experience rather than the individual, and mapped actions, thoughts, and emotions across each stage.

The reason for making this was to take scattered observations like "people follow the crowd," "groups feel random," and "no one asks questions," and place them into a single continuous flow. By doing this, we were able to see how early moments, especially the transition and group formation, directly shape

everything that happens later in the tour. It makes the cause-and-effect relationship visible instead of treating each issue as isolated.

Critical Points

The first breakdown shows up on the **website**. People have trouble understanding the differences between tour options, so they start the experience without a clear idea of what they signed up for.

The biggest spike in confusion happens right **after the presentation**. People leave at the same time, look for direction, and end up relying on the crowd. Movement is fast, but it is not clear.

Group formation has the biggest impact. Since placement is based on timing and position, people end up in groups that don't match what they care about. That mismatch stays for the entire tour and can't be fixed later.

What happens during the tour comes from those earlier steps. Large groups make it harder to speak, and the guide has to stay general to cover everyone. Most people listen instead of asking questions, and specific interests don't get addressed.

Evaluation

Heuristic Review

We evaluate where the Purdue tour breaks basic usability principles like clarity, guidance, and user control based on how the system actually runs, because our observations showed that confusion and mismatch are not isolated issues but repeated breakdowns at specific touchpoints during the transition and group formation process.

- **User Control**
 - **Touchpoint:** Group formation after the presentation
 - **Problem:** Visitors are not given a way to choose a group based on their goals. staff pull people into groups based on position and timing, not interest.
 - **Impact:** High. removes agency at the most critical moment and guarantees that group assignment does not reflect what the visitor wants.
- **Clarity of Next Step**
 - **Touchpoint:** Exit from presentation → outdoor/lobby space
 - **Problem:** When the presentation ends, there is no clear, dominant cue telling visitors exactly what to do next other than the student tour guides yelling trying to organize the tours. Instructions do not carry into the physical space strongly enough.
 - **Impact:** High. visitors hesitate, scan the environment, and default to whatever action seems most obvious rather than following a clear instruction.
- **Visibility of Guidance**
 - **Touchpoint:** Line formation and early movement
 - **Problem:** Any available cues (staff, signs) are weaker than the crowd itself. Once people start moving, visibility of guidance drops and the crowd becomes the main signal.
 - **Impact:** High. navigation becomes social. visitors follow whoever moves first or whichever group looks formed, rather than making an intentional choice.
- **Information Relevance**
 - **Touchpoint:** Tour guide delivery and route
 - **Problem:** Groups are mixed and the route is fixed, so guides have to stay broad. visitors with specific goals (major, labs, identity-based questions) do not get targeted information.
 - **Impact:** High. engagement drops, most people stay passive, and important questions remain unanswered.

Cognitive Walkthrough

To evaluate the system's learnability, we simulated the experience of a first generation, out of state prospective student with high information stakes, meaning they have invested time and money to visit but have no prior familiarity with Purdue, and whose goal is to identify and join a tour group that aligns with their academic interests (e.g., Engineering) immediately after the presentation.

Step	Intended Action	Analysis & Failure Points	Results
1	Exit the Building	Will they know to do this? Yes. The presentation ends and visitors are directed to the exit. Will they see the right cue? No. The exit is congested, and there is no "Next Step" sign. Only that some tour guides are organizing the tours.	Failure: The user pauses at the exit or follows the wrong crowd.
2	Identify the Group	Will they find it easily? Perhaps. Student tour guides are splitting people up into random groups, and it's not a great process due to the amount of people there are.	Failure: The user wanders around, asks others, or waits unsure of what to do. Some may never realize groups are forming until it's too late.
3	Join the Line	Will they join a group that fits their interest? No. Groups are formed completely randomly. Not interest or goals Will they feel confident in their choice? No, there's no information about tour content, guide or area to help them decide.	Failure: The user joins the first group they see or follows others, with no way to know if the tour addresses their academic or personal interests.
4	Verify Progress	Will they see that they are on track? No. There is no preview of the tour route or topics, and no opportunity to switch once in the group.	Failure: The user is locked into a general tour that may not answer their key questions or show what they hoped to see.

Walkthrough Findings:

1. **Passive Assignment**

There is no moment where the visitor actively chooses a group based on their goals. Groups are formed based on timing and position, not intent. Visitors enter with specific interests (major, housing, student life), but the system does not allow them to act on those. This removes agency and turns group assignment into a passive outcome rather than a decision.

2. **Crowd Driven Navigation**

When the presentation ends, movement starts all at once and cues are weak. Signs and staff are not strong enough to guide behavior at scale. Visitors rely on the most visible signal available: other people. They follow whoever moves first or whichever group looks established. Navigation becomes social rather than intentional.

3. **Early Loss of Relevance**

Because all tours follow the same general route and mixed groups are formed randomly, individual goals are not carried into the experience. The issue is not being in the “wrong” group, but that no group is tailored. The system standardizes the experience too early, forcing visitors with different goals into the same generic path.

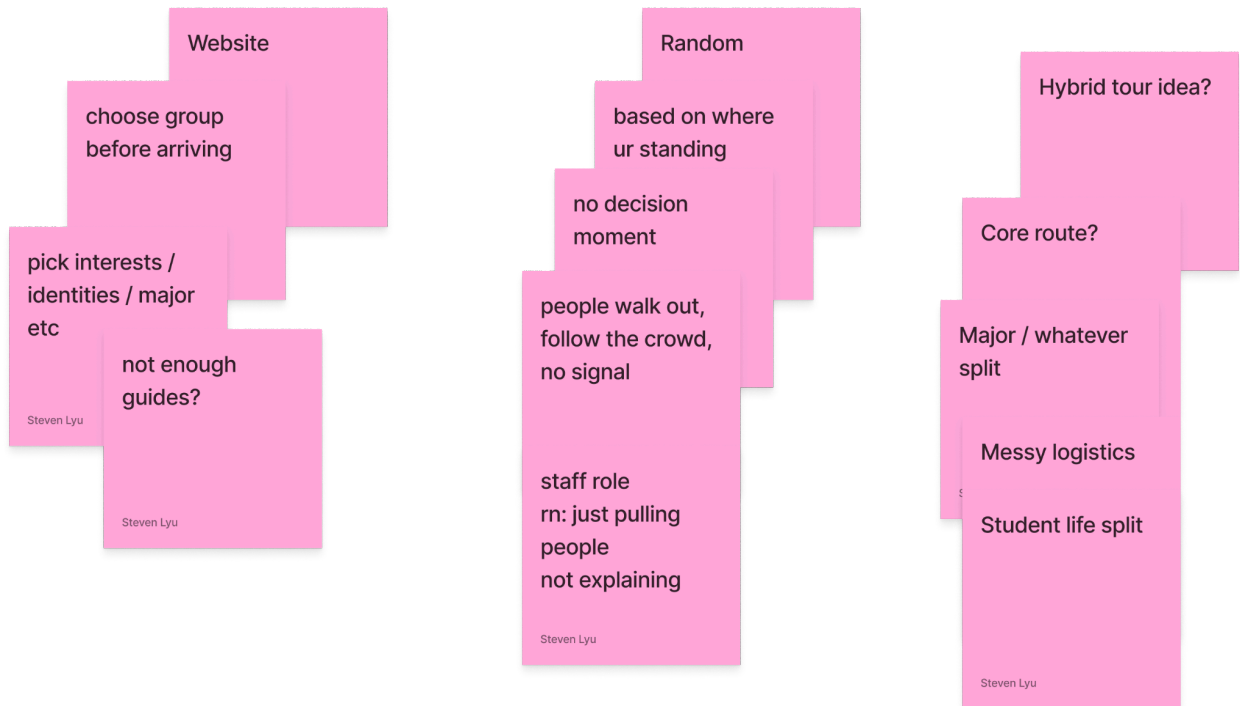
Usability Issues

The system fails because it prioritizes moving large groups efficiently over helping visitors act on their goals. After the presentation, there is no clear structure guiding how groups form, so people rely on proximity and crowd movement rather than making a deliberate choice.

Because grouping is random and not tied to interest, visitors have no control over how their experience aligns with what they came to learn. This removes agency at the most important moment and turns group formation into a passive outcome.

As a result, the tour becomes generic by default. Different goals (major, housing, student life) are flattened into the same experience, forcing guides to stay broad and limiting how useful the tour feels for individuals.

Envisioning



Current Ideas

We started by looking at the moment right after the presentation. People don't really choose anything there. They hesitate, then follow others. Grouping ends up based on timing and position instead of what they actually want.

We explored a more structured way to form tour groups instead of relying on random, on the spot assignments. The main idea starts before the tour, where visitors can quickly indicate their actual college (E.g. College of Agriculture; Daniels School of Business, etc.). Based on this, visitors are assigned to a group ahead of time and given a clear identifier that stays consistent throughout the experience. This identifier system is physical and easy to recognize: each visitor receives something like a color-coded badge or card, which directly matches a tour guide's flag or sign and a designated waiting zone. All three elements (visitor, guide, space) use the same label, so people can immediately understand where they belong without needing to interpret instructions or follow the crowd.

To support this, instructions are communicated early, before the transition begins, so visitors already know how to find their group when the presentation ends. The goal is to remove hesitation at that critical moment and replace guesswork with clear, visible cues. This approach focuses on fixing the transition and

group formation stages without changing the entire tour structure. Instead of redesigning routes, we keep the tour itself simple and consistent, while making sure people start in the right place.

We also considered lightweight support systems around the tour. A followup system can send personalized information after the visit based on what visitors said they were interested in, helping fill in gaps that the tour may not cover. On the guide side, simple training protocols can encourage more interaction, such as prompting questions or creating moments for discussion, to improve engagement within each group.

At the same time, we ruled out several more complex directions. Hybrid or modular routes were dropped because they are difficult to coordinate and can cause groups to fall out of sync. Segmented grouping based on identity factors was avoided due to sensitivity concerns and the risk of making visitors uncomfortable. Major specific tours were also dropped due to there being just way too many majors. Fully dynamic or highly flexible systems were also avoided because they add too much logistical complexity at scale. The final direction stays focused on clarity, consistency, and low effort coordination, using simple physical and procedural changes to reduce confusion and improve alignment.

Core system

- Pre-tour input of the prospective students' college.
- Pre-assigned groups instead of on-the-spot sorting
- Persistent identifier carried through the entire experience
- Matching system across visitor badge, guide identifier, and waiting zone
- Early instructions given before the presentation ends

Physical

- Visitor identification through color badges, clip cards, wristbands, or write-in cards
- Guide identification using tall flags, matching shirts with icon, color, and text, plus voice cues
- Waiting zones using floor lanes, separated spatial areas, and strong visual anchors
- Instructional signage repeated across exit points, hallways, and outdoor spaces

Flow control

- Guided release of groups instead of everyone exiting at once
- Walking trails using continuous color paths beyond the exit
- Live boards or displays showing group directions and updates

Engagement support

- Guide prompting strategies to create moments for questions
- Question tokens to lower pressure for participation
- Prompt cards to help visitors know what to ask

What stays the same

- Overall tour structure
- No reliance on complex digital systems

What we avoided

- Modular or hybrid routes due to coordination issues
- Major specific tours due to scale limits
- Identity based grouping due to sensitivity concerns
- Fully dynamic systems due to high logistical complexity
- Follow up + physical info sheet due to a lack of time

Final direction

- Move complexity before the tour through assignment and labeling
- Keep execution simple during the tour through clear signals and controlled flow
- Prioritize clarity somewhat more over flexibility

Prototyping (Physical)

We translated these ideas into a set of physical prototypes to test how simple, visible signals can guide visitors through group formation without relying on guesswork. The following is a list of ideas that the Purdue Tour could use to improve the experience of new coming students. These are not meant to be implemented all at once, but rather as a range of options that can be combined or tested individually depending on feasibility and context.

① Visitor Identifier

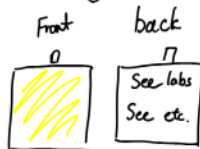
- Color badges with labels



- Clip-on cards
small colored card clipped to shirt



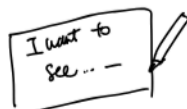
- Double-Sided badge
Front - color
back - goal



- Wristband with info
Scan for visitor info
match color with badge



- Write in cards



② Guide Identifiers

- Tall Flags
Large colored flag above the head



- Voice Quas
Guide repeating phrases
working mic (!!!)



- Icon symbol + color + Text



- Guides Line up outside
In prepositioned zones



(See Appendix for the PDF of our low-fi sketches)

Visitor Identification

People were just following others, some tag along

Goal: Make visitor intent visible early so grouping isn't random.

- **Color badges**
 - Easiest way to show group from a distance
 - Removes need to ask or think
- **Clip on cards**
 - Faster than lanyards at check-in
 - Less setup, easier to distribute
- **Double sided badges**
 - Front = quick group signal
 - Back = more specific interest
 - Helps guides adjust without changing grouping
- **Wristbands**
 - Harder to lose in movement
 - Easier to scan in large crowds
- **Write in cards**
 - Note down information that the students might find an interest in during the presentation
 - Note down locations that they want to see

Guide Identification

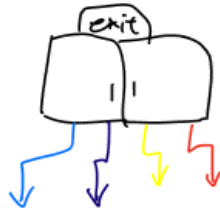
Visitors couldn't find or distinguish guides during transition

Goal: Change guessing into knowing, stronger signals

- **Tall flags**
 - Visible above crowd
 - Not blocked like normal signs
- **Voice cues**
 - Backup to visual signals
 - Helps in noisy, crowded movement
- **Icon + color + text on shirts**
 - Matches how visitors think (engineering, business, etc.)
 - Removes need to translate abstract labels
- **Pre positioned guide lines**
 - Guides already in place before people exit

③ Waiting Zones

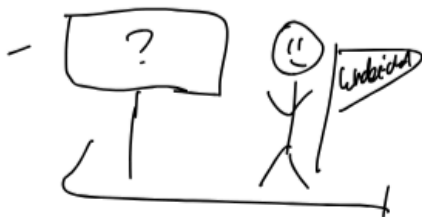
- Floor lanes
Colored tape from exit



- Spatial Zones
groups for content



- Visual Anchors



④ Instructional Signage

- Pre-exit signs
instructions before leaving
the room

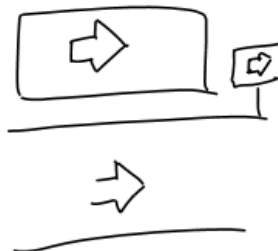


- Repeated Signs

same sign repeated along the path



- More Signs !



Waiting Zones

People exit and immediately hesitate with no clear path, so they follow the crowd

Goal: Replace wandering and crowd following with direct movement

- **Floor lanes**
 - Colored paths starting at the exit
 - Each lane represents one group
 - Removes the need to decide, just follow your color
- **Spatial zones**
 - Clearly separated areas for each group
 - Physically spaced apart, not just visual
 - Prevents groups from mixing
- **Visual anchors**
 - Strong markers that tie everything together
 - Same color across ground, sign, and guide
 - Gives people something stable to move toward

Instructional Signage

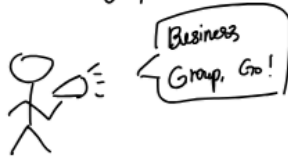
Current signs exist but do not control behavior, so people ignore them and follow others

Goal: make the instruction unavoidable and consistent so people do not rely on guessing

- **Pre existing signs**
 - Instructions shown before people move
 - Sets expectation early
- **Repeated signs**
 - Same message placed multiple times
 - Exit, hallway, and outside
 - Reduces missed information
- **More signs**
 - Wall signs, floor arrows, and standing boards
 - All showing the same message
 - Ensures visibility even in crowds

System Context

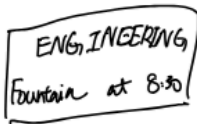
- ① Guided Release
call groups one by one



- ② Walking trails



- ③ Live board updates



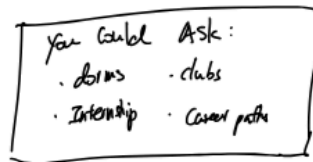
Engagement Tools

- ① Question-tokens
physical tokens for asking



- ② Prompt Cards

Cards with suggested questions



(See Appendix for the PDF of our low-fi sketches)

System Control

People all move at once after the presentation, which creates crowding and confusion

Goal: control movement so the system stays clear instead of breaking under crowd behavior

- **Guided release**
 - Call groups one at a time instead of everyone leaving together
 - Reduces crowd pressure at the exit
 - Gives each group a clear moment to move
- **Walking trails**
 - Colored paths that continue beyond the exit into the tour route
 - Keeps groups organized after initial formation
 - Prevents groups from blending back together
- **Live board updates**
 - Whiteboards or digital displays showing real-time info
 - Helps with live updates of groups, use in place of audio cues.

Engagement Tools

Most visitors stay passive and avoid asking questions during the tour

Goal: make interaction easier so visitors do more than just listen

- **Question tokens**
 - Each visitor gets a limited number of tokens
 - Used to ask questions during the tour
 - Encourages participation without forcing it
- **Prompt cards**
 - Cards with suggested questions or topics
 - Helps visitors who do not know what to ask
 - Reduces hesitation and lowers social pressure

Prototyping (Digital)

VISIT PURDUE



The best way to imagine your future at Purdue is to experience it for yourself. Visits allow you to explore Purdue's locations and interact with the people who set us apart. Learn about academic programs and student life, take an in-person walking tour and connect with our community.

Listed below are the tour options Purdue provides for prospective students. We look forward to seeing you!

Prospective Student Tour

[Book a tour >>](#)

OR

Middle/High School Tours

[Book a tour >>](#)

Self-guided Tour:

Download the Purdue Guide app for an audio tour or print a copy of our West Lafayette or Indianapolis maps to take a self-guided walking tour. Use the links to take you to the app's download page:

[Google Play Store >>](#)
[App Store >>](#)

Getting to Purdue:

Plan your visit to Purdue with these helpful travel tips, including directions, parking, and lodging.

[Learn More](#)

Here is our prototype for the redesigned website. We specifically designed the webpage to be a little more clear on where the users should be clicking if they are interested in a tour.

Section 2

PROSPECTIVE STUDENT TOUR



Explore Purdue in person or virtually to discover what it means to be a Boilermaker. Our prospective student visit experiences include options in West Lafayette, Indianapolis, in your region and online. You can go on a tour, chat with current students and meet with your admissions counselor. Take your next small step by joining us for one of our visit opportunities.

The Prospective Student Tour consists of three main segments: the **Introduction**, the **Academic Tour**, and the **Residence/ Dining Tour**.

Tour Schedule + Booking:

1. After you've parked at the Grant Street Parking Garage (located near the PMU), make your way to the Boilermaker Welcome Center, located in Room 102 of the Stewart Center.
2. After checking in, you will be presented a general intro to Purdue, located in the Stewart Center's first floor Theater.
3. Follow the guide outside and get into your tour groups for the **Academic Tour**! If you would like to participate in the Academic Tour, please **pick your department/school from the drop-down**:

Select School ▾
4. After the Academic Tour, you will be lead by your tour guide to the Engineering Fountain for the **Dining and Residence Tours**, which you will have a choice of picking between. You can also schedule either individually below if you would like to do both or simply would prefer to do them a different time!

Choose from below which tour you'd prefer. (You can change your mind on the day of the tour, this is just helpful for planning ahead.)

I'd prefer the Residence Tour

I'd prefer the Dining Tour

Book a Prospective Student Tour >>

Read a description about each of the tours below, and schedule a tour individually if you'd like!

Here, because we want the users to choose the school / college that they were accepted into, there is a select school button which leads to a dropdown menu of all the colleges that Purdue has to offer. For example: College of Agriculture; Daniels School of Business; College of Education; etc.

Read a description about each of the tours below, and schedule a tour individually if you'd like!

Academic Tour:

Tour through Purdue's academic campus, getting shown a lay of the land and how students go about their day-to-day academic life, while also getting a department-specific tour of the academic building(s) that will be most important to your program.

[Book a tour >>](#)

Dining Tour:

Take a tour across Purdue's many dining courts and retail dining locations. Learn about what dining with Purdue Dining & Culinary looks like and about the different dining plans available to students.

[Book a tour >>](#)

Residence Tour:

Take a tour through one of Purdue's residence halls and get an idea what living in a Purdue dormitory would feel like. Learn about the services students have as they live at Purdue and get a tour of the overall residence area and what's around.

[Book a tour >>](#)

Here is what would happen if you scroll down on the website. It gives a brief little insight into what each tour is like.

Section 3
2026
March

SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
Available	Available	Fully booked	Fully booked	Available	Available	Available
Available	Fully booked	Available	Available	Available	No Tour Scheduled	Available
No Tour Scheduled	Fully booked	Available	Available	Fully booked	No Tour Scheduled	Available
Available	Available	Fully booked	Available	Available	Available	Fully booked
No Tour Scheduled	Available	Available	Available	Available	Available	Fully booked
Fully booked	No Tour Scheduled	Available	Available	Fully booked	Available	Available

Available

Fully booked

No Tour Scheduled

Available tour time for selected day - XX:XX

Provide email:

Book Tour!

A confirmation email will be sent to you.
Please have you confirmation email ready to show when you check-in.

This small page is meant to demonstrate the pop-up that appears over the screen when any “Book Tour” button is clicked. It is a simple calendar that shows you what days are available to schedule tours and what the tour time will be. Then the user is asked for their email, and it is communicated to them that this is for a confirmation email that they’ll use to check in. Then, they are able to book the tour.

Usability Testing

We did not run a formal usability test with structured tasks, metrics, or controlled conditions. Instead, we did a quick informal pass with a small number of people to see how they react to the ideas.

We selected three participants and walked them through our concepts, mainly focusing on the physical signaling system (badges, guide identifiers, and waiting zones). We explained the scenario and asked them how they would move through the space and what they would expect to do at each step.

The goal was not to measure performance, but to see where confusion still exists and whether the system feels understandable at a glance.

What we did

- Explained the current problem (random grouping, crowd following)
- Showed the proposed system (color badges, guide flags, zones)
- Asked them to describe what they would do after the presentation ends
- Asked what felt clear vs unclear
- Asked what they would change

What we observed

- All three understood the color matching idea quickly
- They expected the system to work only if it is consistent across all elements (badge, flag, space)
- They pointed out that instructions need to be given before the transition, not during
- They said they would still follow the crowd if signals are weak or partially visible
- They preferred fewer, stronger signals over many competing ones

Suggestions from participants

- Make the colors and labels very simple
- Add one dominant instruction at the end of the presentation
- Ensure guides are visible before people exit
- Avoid overloading with too many signs or options
- Perhaps add a physical booklet for all the information

Limitations

Logistical

- Limited number of tour guides makes grouping hard to scale without delays or uneven group sizes
- Pre-assignment requires coordination before arrival and depends on visitors actually completing preferences
- Physical signaling requires setup, staffing, and consistent enforcement

Reliability

- Color coded or visual systems can break under crowd density
- Visitors may ignore or misunderstand labels even if improved
- Dependence on staff execution

User Behavior

- Large groups will still reduce participation regardless of grouping logic
- Visitors may not engage with the tools that we provide them

Tradeoff

- More structure = better clarity but less flexibility in movement
- Faster throughput systems conflict with personalized experiences
- More signals can reduce confusion but also increase cognitive load if overdone

Scope

- Findings are based on a limited sample of interviews and observations, not full population data
- Behavior observed in one campus context may not generalize to all universities
- Some conclusions rely on inferred behavior patterns (e.g., crowd-following) rather than controlled testing
- Prototypes and ideas have not been fully validated in real world situations

Implementation

- Requires coordination between admissions, guides, and infrastructure changes
- Training burden increases (guides must adapt to new systems and possibly tailored tours)
- Physical redesign may be constrained by space and campus layout
- Incremental rollout is difficult

Conclusion

This project focused on how visitors are grouped during campus tours and how that moment shapes everything that follows. The current process moves people quickly but relies on timing and crowd behavior, which leads to confusion and mismatched groups. Once visitors are placed, the rest of the experience is largely fixed. Engagement, relevance, and overall satisfaction are all tied back to that early step.

The direction we developed centers on making group assignments clear before movement begins. Simple identifiers, visible guides, and defined zones aim to remove hesitation and reduce reliance on following others. The goal is to let visitors understand where they belong immediately, so the tour starts with alignment instead of randomness.

Because the timeline was tight, and we did not start as early as planned, we compressed most of the work into a short period. This affected how much we could do. Research stayed limited in scale, and testing remained informal without real crowd conditions. Prototypes were kept at a low level and were not validated in live environments.

Some ideas were not fully developed due to time. We intended to create a dedicated take home piece, such as a PDF or pamphlet, that organizes tour information based on visitor interests. This would give visitors a way to revisit what mattered to them after the tour, instead of relying on memory or searching later. That direction was left incomplete.

The concepts presented show a clear direction for improving clarity and alignment during group formation, but they remain untested at scale. Further work would require real world implementation, larger participant groups, and coordination with existing tour operations to evaluate how these changes perform under actual conditions.

Contributions

- Introduction	
- Campus Tours at Purdue	Sylas / Jeric / Shaishav / Alvin
- The Problem	Sylas / Jeric / Shaishav / Alvin
- Contextual Research	
- Questions	Sylas / Jeric / Shaishav / Alvin
- Tours (Website Specific)	Shaishav
- Tours	Sylas / Jeric
- Secondary Research	
- Questions	Sylas / Jeric
- Results	Sylas / Jeric / Shaishav / Alvin
- Synthesis	
- Journey Mapping	Sylas
- Breakdown Goals	Sylas
- Prioritize	Sylas
- Analysis	
- Task Analysis	Jeric / Sylas
- Critical Points	Jeric / Sylas
- Evaluation	
- Heuristic Review	Alvin / Sylas
- Cognitive Walkthrough	Alvin / Sylas
- Walkthrough Findings	Alvin / Sylas
- Usability Issues	Alvin / Sylas
- Envisioning	Sylas / Jeric / Shaishav / Alvin
- Prototyping (Physical)	Sylas
- Prototyping (Digital)	Shaishav
- Usability Testing	Sylas
- Limitations	Sylas
- Conclusion	Sylas

Appendix

[JPG](#) for Journey Map

[PDF](#) for Physical Prototypes

[Figma](#) for Digital Prototype